

Probability and Bayesian Reasoning

Lecture 18

CS 4880

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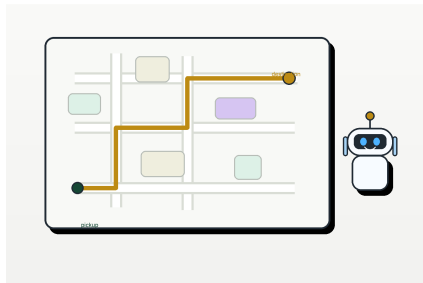
The Robot Already Knows How to Get There

A campus delivery robot has already computed a route with a path-finding algorithm.

- ▶ The map contains sidewalks, crossings, ramps, and buildings.
- ▶ The planner chooses a low-cost path to the destination.
- ▶ On the static map, the route is valid.

Key idea

Search can solve the route-planning problem perfectly relative to the map it was given.

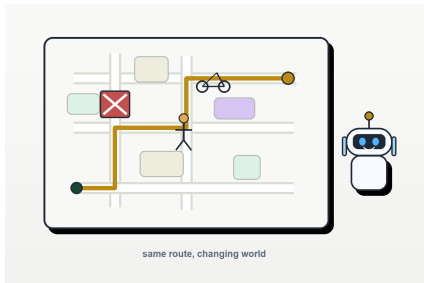


But the Campus Is Not Static

Between planning and execution, the route may change:

- ▶ pedestrians stop in the walkway;
- ▶ bicycles cross the robot's path;
- ▶ maintenance carts appear;
- ▶ construction barriers move;
- ▶ temporary events block a sidewalk.

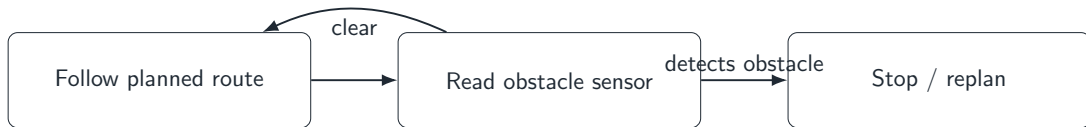
The map describes what *should* be there. The robot must reason about what is there *now*.



The Contingency Plan: Detect, Stop, Replan

A reasonable controller might contain a rule like:

if obstacle detected $\Rightarrow$ stop and replan.



Key idea

The planner is no longer the hard part. The hard part is deciding whether the sensor reading should be believed.

Can the Sensor Ever Be Trusted 100%?

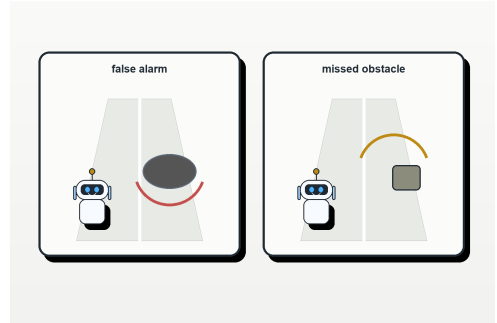
Even a very good detector can produce:

False positive

The sensor reports an obstacle when the path is actually clear.

False negative

The sensor reports clear when an obstacle is actually present.



Key idea

Once perception can be wrong, the robot needs a language for uncertainty rather than only true/false rules.

Random Variables for the Robot

Define two binary random variables:

$$O \in \{\text{obstacle}, \neg\text{obstacle}\}$$

$$S \in \{\text{sensor+}, \text{sensor-}\}.$$

We will abbreviate the events as

$$O, \neg O, S, \neg S.$$

Definition

A random variable represents an uncertain quantity whose possible values are known even when its current value is not.

Key idea

The obstacle either exists or does not. The uncertainty belongs to the robot's information state.

A Probability Distribution Assigns Belief Mass

Suppose the robot's recent campus data suggests

$$P(O) = 0.10.$$

Then, because O is binary,

$$P(\neg O) = 1 - P(O) = 0.90.$$

More generally, a valid distribution satisfies

$$0 \leq P(x) \leq 1 \quad \text{and} \quad \sum_x P(x) = 1.$$

Key idea

Before the sensor reports anything, the robot already has a **prior** belief about how likely the path is to be blocked.

The Prior Comes from Context

The value $P(O) = 0.10$ is not a universal constant. It could depend on:

- ▶ time of day;
- ▶ location on campus;
- ▶ weather;
- ▶ class-change periods;
- ▶ construction schedules;
- ▶ the robot's recent observations.



Key idea

Probability models encode assumptions about the environment. A wrong prior can distort even mathematically correct inference.

The Sensor Model Is Conditional

Suppose calibration gives the detector the following behavior:

$$P(S \mid O) = 0.90$$

$$P(S \mid \neg O) = 0.05.$$

Therefore

$$P(\neg S \mid O) = 0.10, \quad P(\neg S \mid \neg O) = 0.95.$$

Sensitivity / true-positive behavior

$P(S \mid O)$ asks how often the sensor fires when an obstacle really exists.

False-positive behavior

$P(S \mid \neg O)$ asks how often it fires on a clear path.

From Conditionals to Joint Probabilities

The product rule gives

$$P(A, B) = P(A)P(B \mid A).$$

For the robot:

$$P(O, S) = P(O)P(S \mid O) = 0.10(0.90) = 0.090$$

$$P(O, \neg S) = 0.10(0.10) = 0.010$$

$$P(\neg O, S) = 0.90(0.05) = 0.045$$

$$P(\neg O, \neg S) = 0.90(0.95) = 0.855.$$

Key idea

A joint probability refers to a complete combination of events: what the world is like *and* what the sensor reports.

How Often Does the Sensor Fire at All?

The sensor can fire in two mutually exclusive worlds:

1. an obstacle exists and the sensor fires;
2. no obstacle exists and the sensor fires anyway.

Therefore

$$P(S) = P(O, S) + P(\neg O, S)$$

$$P(S) = 0.090 + 0.045 = 0.135.$$

Likewise,

$$P(\neg S) = 0.010 + 0.855 = 0.865.$$

Key idea

Marginalization sums over hidden alternatives that could have produced the observation.

The Direction We Know Is Not the Direction We Need

Calibration gave us

$$P(S | O) = 0.90.$$

But after the robot observes a positive sensor reading, the operational question is

$$P(O | S) = ?$$

These are not the same quantity.

$$P(S | O)$$

If there is an obstacle, how likely is the sensor to fire?

$$P(O | S)$$

If the sensor fired, how likely is there actually an obstacle?

Conditional Probability Narrows the Sample Space

By definition,

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

when $P(B) > 0$.

For our robot,

$$P(O \mid S) = \frac{P(O, S)}{P(S)}.$$

We already computed both terms:

$$P(O, S) = 0.090, \quad P(S) = 0.135.$$

Key idea

Conditioning means restricting attention to the worlds consistent with the evidence and renormalizing their probability mass.

Bayes' Rule Comes from the Product Rule

The same joint event can be written in two directions:

$$P(A, B) = P(A)P(B | A)$$

and

$$P(A, B) = P(B)P(A | B).$$

Set them equal:

$$P(A)P(B | A) = P(B)P(A | B).$$

Solve for $P(A | B)$:

$$\boxed{P(A | B) = \frac{P(B | A)P(A)}{P(B)}}.$$

Key idea

Bayes' rule reverses a conditional probability by combining a likelihood with a prior.

Worked Example: Sensor Fires

We know

$$P(O) = 0.10, \quad P(S | O) = 0.90, \quad P(S) = 0.135.$$

Apply Bayes' rule:

$$\begin{aligned} P(O | S) &= \frac{P(S | O)P(O)}{P(S)} \\ &= \frac{0.90(0.10)}{0.135} = \frac{0.090}{0.135} \approx \boxed{0.667}. \end{aligned}$$

Key idea

A positive sensor reading raises the obstacle probability from 10% to about 67%—strong evidence, but not certainty.

Why Doesn't a 90% Detector Give 90% Confidence?

Among 1,000 comparable path segments, the model predicts approximately:

	Obstacle	Clear
Sensor+	90	45
Sensor-	10	855

When the sensor fires, there are $90 + 45 = 135$ positive readings.

Only 90 correspond to real obstacles:

$$\frac{90}{135} = 0.667.$$

Key idea

The posterior depends on both sensor quality and the base rate of obstacles.

Negative Evidence Also Updates Belief

Suppose the sensor reports clear. Then

$$P(O \mid \neg S) = \frac{P(\neg S \mid O)P(O)}{P(\neg S)}.$$

Substitute the model values:

$$P(O \mid \neg S) = \frac{0.10(0.10)}{0.865} = \frac{0.010}{0.865} \approx \boxed{0.0116}.$$

The robot's belief in an obstacle falls from 10% to about 1.2%.

Key idea

Evidence can increase or decrease a belief; Bayesian updating is not inherently “alarm seeking.”

Probability Is Not Yet a Decision Rule

After a positive reading, we found

$$P(O \mid S) \approx 0.667.$$

Should the robot stop?

- ▶ Stopping unnecessarily costs time and battery.
- ▶ Continuing into a real obstacle can damage the robot or injure someone.
- ▶ Replanning may be cheap in one location and expensive in another.



Key idea

Probability tells the agent what to believe. Choosing an action also requires costs, preferences, or utility—a later topic.

What If the Robot Samples Again?

Suppose the robot pauses briefly and obtains a second positive reading S_2 after the first S_1 .
We want

$$P(O \mid S_1, S_2).$$

A useful decomposition is

$$P(O \mid S_1, S_2) \propto P(S_1, S_2 \mid O)P(O).$$

If the readings are conditionally independent given the true obstacle state,

$$P(S_1, S_2 \mid O) = P(S_1 \mid O)P(S_2 \mid O).$$

Key idea

Multiple evidence can strengthen a belief, but only if we model how the evidence pieces relate to one another.

Two Positive Readings: Update by Relative Weight

Under the conditional-independence assumption:

$$w_O = P(O)P(S_1 | O)P(S_2 | O) = 0.10(0.90)(0.90) = 0.081$$

$$w_{\neg O} = P(\neg O)P(S_1 | \neg O)P(S_2 | \neg O) = 0.90(0.05)(0.05) = 0.00225.$$

Normalize:

$$P(O | S_1, S_2) = \frac{0.081}{0.081 + 0.00225} \approx \boxed{0.973}.$$

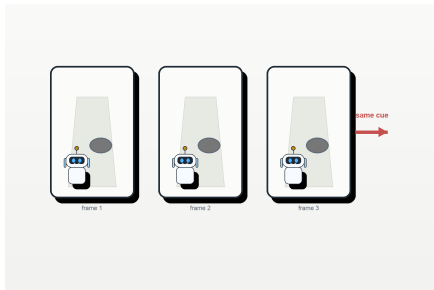
Key idea

Two apparently independent positive readings move the belief from 10% prior to 67% after one reading to about 97% after two.

But Are Two Readings Really Independent?

Repeated measurements from the same sensor may share the same failure cause:

- ▶ glare remains in the camera;
- ▶ a reflection persists across frames;
- ▶ a dirty lens affects every observation;
- ▶ the same occlusion hides the same obstacle;
- ▶ the same learned detector bias repeats.



Key idea

Treating correlated evidence as independent can make the posterior dramatically overconfident.

Independence Is a Modeling Claim

Two events A and B are independent when

$$P(A, B) = P(A)P(B).$$

Equivalently, when $P(B) > 0$,

$$P(A \mid B) = P(A).$$

Conditional independence is different:

$$P(A, B \mid C) = P(A \mid C)P(B \mid C).$$

For repeated sensor readings we might assume

$$S_1 \perp S_2 \mid O.$$

Key idea

Independence is not something probability gives us for free. It is an assumption that simplifies the model.

Normalization Turns Scores into Probabilities

Often we compute only proportional scores:

$$P(O \mid e) \propto P(e \mid O)P(O).$$

For binary O , define

$$w_O = P(e \mid O)P(O), \quad w_{\neg O} = P(e \mid \neg O)P(\neg O).$$

Then

$$P(O \mid e) = \frac{w_O}{w_O + w_{\neg O}}.$$

Or write

$$P(O \mid e) = \alpha P(e \mid O)P(O),$$

where α is the normalizing constant that makes the probabilities sum to 1.

Many Evidence Variables Become Expensive

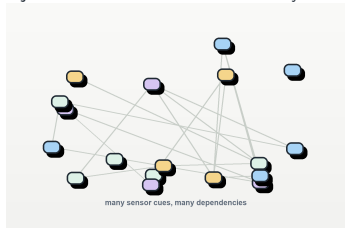
Suppose the perception system produces evidence

$$E_1, E_2, \dots, E_n.$$

In general, the robot would need

$$P(E_1, E_2, \dots, E_n \mid O),$$

a joint model that accounts for every interaction among the evidence variables.



Key idea

Exact modeling becomes expensive because combinations of uncertain variables grow rapidly.

Naive Bayes Makes a Deliberately Strong Assumption

Naive Bayes assumes that evidence variables are conditionally independent given the class:

$$E_i \perp E_j \mid C.$$

Then

$$P(C \mid E_1, \dots, E_n) \propto P(C) \prod_{i=1}^n P(E_i \mid C).$$

For obstacle detection:

$$P(O \mid E_1, \dots, E_n) \propto P(O) \prod_i P(E_i \mid O).$$

Key idea

The assumption is “naive” because the evidence is often not truly independent—but the simplification makes inference extremely cheap.

Naive Bayes from One Sensor's Features

A single perception sensor can still produce several features:

- ▶ E_1 : object-like shape detected;
- ▶ E_2 : path region is visually occupied;
- ▶ E_3 : detection persists across a short interval.

Suppose

$$P(E_1 \mid O) = 0.85, \quad P(E_2 \mid O) = 0.80, \quad P(E_3 \mid O) = 0.75$$

and

$$P(E_1 \mid \neg O) = 0.08, \quad P(E_2 \mid \neg O) = 0.10, \quad P(E_3 \mid \neg O) = 0.15.$$

Key idea

Naive Bayes converts a collection of simple feature likelihoods into a posterior over the class of interest.

Worked Naive Bayes Classification

With prior $P(O) = 0.10$:

$$w_O = 0.10(0.85)(0.80)(0.75) = 0.051$$

$$w_{\neg O} = 0.90(0.08)(0.10)(0.15) = 0.00108.$$

Normalize:

$$P(O \mid E_1, E_2, E_3) = \frac{0.051}{0.051 + 0.00108} \approx \boxed{0.979}.$$

Key idea

The calculation is fast because we multiply small local likelihoods instead of storing one enormous joint distribution.

The Engineering Tradeoff: Simplicity vs. Calibration

What the simplifying assumptions buy us

- compact probability tables
- fast updates
- easy estimation from data
- interpretable likelihood terms

What they can cost us

- missed dependencies
- overconfident posteriors
- sensitivity to biased training/calibration data
- systematic errors may be counted repeatedly

Key idea

A probability value can be numerically precise while the model that produced it is structurally wrong.

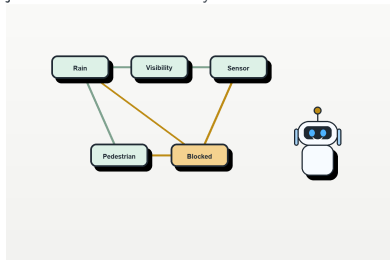
One Robot Quickly Becomes Many Random Variables

A realistic delivery robot may reason about

Pedestrian, Bicycle, Rain, Visibility, SensorReading, BlockedPath, . . .

The challenge is no longer merely applying Bayes' rule once.

We need to represent which variables directly influence which others so that we do *not* store one full joint distribution over every combination.



Key idea

Next: Bayesian networks use graph structure to encode conditional independence and factor a large joint distribution.

Takeaways

- ▶ Random variables represent uncertain aspects of the world and the robot's observations.
- ▶ Priors encode what the agent believed before new evidence arrived.
- ▶ Conditional probabilities model sensor behavior under different world states.
- ▶ Bayes' rule converts likelihoods and priors into posterior beliefs.
- ▶ Multiple evidence can strengthen inference, but dependence matters.
- ▶ Normalization turns relative Bayesian scores into a valid probability distribution.
- ▶ Naive Bayes gains computational simplicity by assuming conditional independence.
- ▶ The recurring AI tradeoff is **model fidelity vs. computational tractability**.

Key idea

The robot should not ask “Did the sensor say obstacle?” It should ask “Given what I knew before and what the sensor just reported, how strongly should I now believe an obstacle is present?”